

Stereo Audio Distribution Buffer for Headphones



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Most audio signal sources have only one stereo output, which means they can drive only a single pair of headphones with a resistance of around 32 ohms or a single line of 600 ohms. But sometimes several people are required to connect their headphones to a single

PARTS LIST

Semiconductors:

- IC1 - NE5532 op-amp
T1, T4 - BC337 npn transistor
T2, T3 - BC327 pnp transistor
LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R7-R10,
R19 - 1-kilo-ohm
R3, R4 - 10-kilo-ohm
R5, R6 - 1-mega-ohm
R11-R18 - 100-ohm

Capacitors:

- | | |
|------------|----------------------------------|
| C1, C2, C4 | - 0.33 μ F ceramic disk |
| C3 | - 100 μ F, 16V electrolytic |
| C5 | - 1000 μ F, 25V electrolytic |
| C6, C7 | - 220 μ F, 25V electrolytic |

Miscellaneous:

- CON1-CON6 - Stereo connector, female
- CON7 - DC input connector
- CON8 - USB connector
- SL, S2 - On/off switch

audio signal source—such as for entertainment, e-learning and training, or at home. In these situations, use of powerful loudspeakers is not desirable because other people in the room will get disturbed.

Also, sometimes the signal from one audio output needs to be sent to several power amplifiers located many metres away from each other. In that case, the distribution device should be able to drive long audio lines with resistance between 100 and 600 ohms.

Presented here is a solution to such a requirement that adds a distribution buffer (DB) between the audio output and the headphones. Most headphones have some kind of volume control. Therefore no volume control is added

Test Points

Test point	Details
TP0	0V, GND
TP1	+5V (with USB)
TP2	Left-channel audio signal
TP3	Right-channel audio signal

to the distribution buffer to keep the things simple.

Circuit and working

Fig. 1 shows the circuit of the stereo distribution buffer built to solve practical problems that arise when the signal from a single stereo source should be buffered in order to drive up to four headphones or small high-impedance loudspeakers with resistance higher than 32 ohms.

The circuit has two identical channels. Each channel of the buffer has one input and four outputs. The input stereo signal is applied on connector CON1. The same signal is available on connector CON2, so the second distribution buffer can be connected to that connector.

R1 and R2 are resistors for input protection, typically having a value between 100 ohms and 2 kilo-ohms. Their exact value depends on the parameters of the signal source, the cables, the inputs, etc.

C1 and C2 are used to cut the DC input component. If DC voltage from

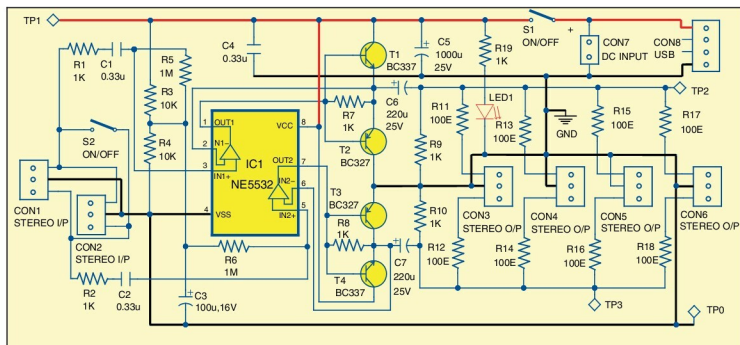


Fig. 1: Circuit of stereo audio distribution buffer for headphones